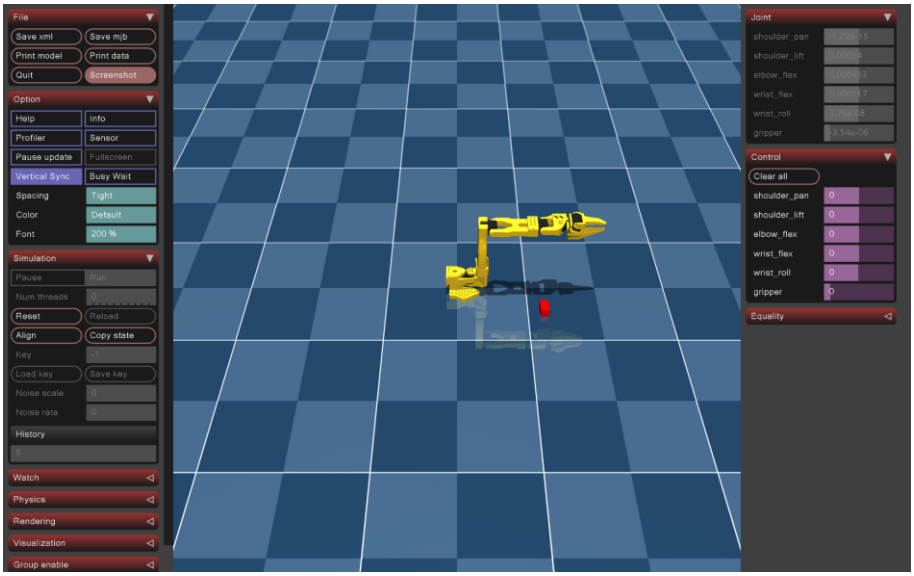
CSE 4360/5364 Autonomous Robotics
Homework 1 – Fall 2026
Due 9/10/2026 11:59PM

1.1 Objectives

The purpose of this lab is to familiarize students with the SO-101 robot arm simulator and make the arm perform a pick and place task.

In this lab, you will:

- Control the SO-101 arm using the Numerical Manipulator Configuration Dialog/Slider to control individual joints
- Demonstrate a sequence of motions such as picking and placing an object



1.2 Task 1: Install Simulation Software

The following instructions are tested on Linux Ubuntu, Windows, and MacOS. If you are using a platform other than the ones listed and are facing any issues, feel free to contact me.

Getting started:

Download the file lerobot-mojuco.zip to a working directory.

Extract the files. The steps to install are present in the README.md. Follow specific instructions for your OS.

1.3 Task 2: Explore

Students will understand how the simulator works and how to control the robot in the simulator.

1.3.1 Procedure:

Run Simulator:

Follow the appropriate command to run the simulator. Two <mode>s is available, explore and command:

- Explore mode: Lets you control the robot with a graphical user interface.
- Command mode: Runs a Python program to control your arm.

Windows:

```
uv run python robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Linux:

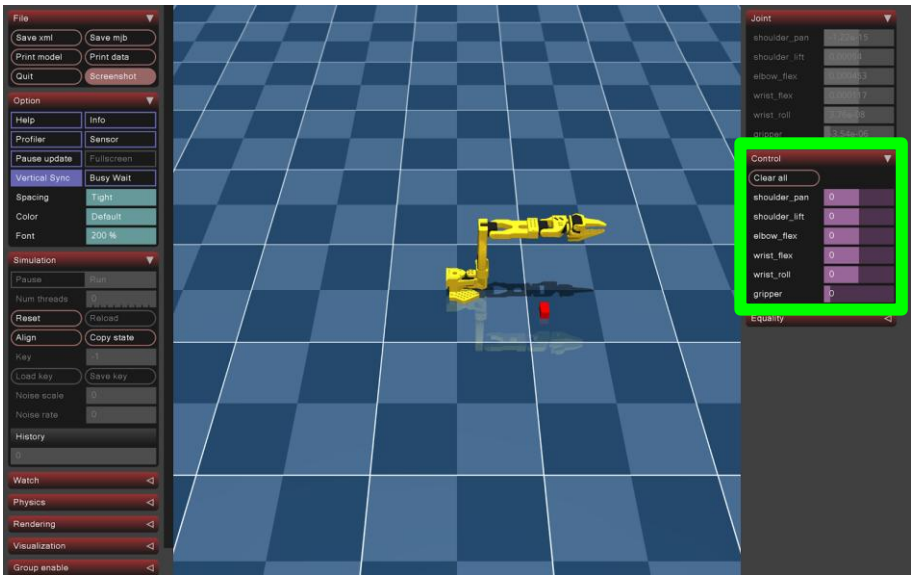
```
uv run python robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Mac:

```
uv run mjpython robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Numerical (Joint) Mode Movement in Explore mode:

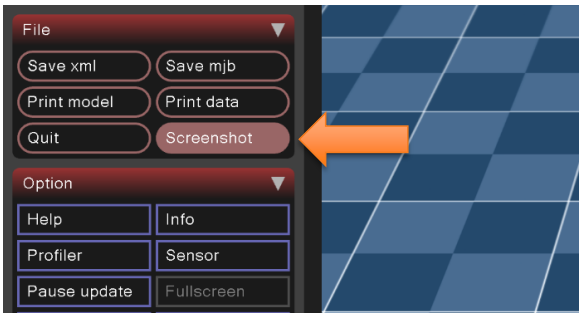
- The right-side panel contains the tab to control the robot. Under the Control tab, all the robot joints are listed and can be controlled with the numerical slider.
- Adjust the slider to see the robot's joints move.
- You can use the mouse to move around the robot.
  - Click and drag to change the camera view
  - Alt+mouse translates
  - Scroll wheel zooms



1.3.2 Reporting

Capture screen shots of the robot in different views:

- Top view
- Side view
- Front view



1.4 Task 3: Place and Pick Program

Students will program the SO-101 arm to grasp the cube and place it in the destination (0.2,0.0,0.0)

1.4.1 Procedure

Run Simulator:

Follow the appropriate command to run the simulator:

There are two modes available the explore or command:

- Explore mode: It lets you control the robot with graphical user interface
- Command mode: Run a python program to control your arm.

Windows:

```
uv run python robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Linux:

```
uv run python robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Mac:

```
uv run mjpython robo-code/sim.py <mode> <last-four-digits-uta-id>
```

Determine joint positions:

- The cube is spawned in a unique location based on the last four digits of your ID number.
  - The current cube location is output to the console.
- Use the Explore mode to find a sequence of joint positions that enables the robot to grasp the cube, then move it to approximately (0.2,0.0,0.0).
  - You will have to write down or save the joint angles as you find them.

Commented [NG1]: This was already given in 1.3?

Commented [TV2R1]: Yes, professor but still wanted to repeat it so they don't have to scroll all the way to the top.

Program the arm:

- A sample function call is demonstrated in the main function.

```
robot.command([
    {"shoulder_pan": 0.0, "shoulder_lift": 0.0,   "elbow_flex": 0.0,
     "wrist_flex": 0.0,  "wrist_roll": 0.0, "gripper": 1.75},
    {"shoulder_pan": 0.0, "shoulder_lift": -0.3,  "elbow_flex": -0.152,
     "wrist_flex": 1.28, "wrist_roll": 0.0, "gripper": 1.75},
], duration=4.0)
```

- `robot.command()` is how you can pass the required joint angels to the robot.
  - `robot.command()` accepts a sequence of joint angels for the robot to execute in a single movement.
  - Items inside the list are formatted as a dictionary.
  - The duration parameter sets how long the robot should take to perform that particular movement.
- Use `robot.command()` to execute the sequences of joint angles that you found earlier on the robot.
- Make sure you don't make any changes to the code above the main function.

1.5 Submission Guidelines

Submit both files as one zip in canvas section. Make sure you follow the specified guidelines for summiting. The zip file should be named as <lastname>_<firstname>.zip

1.5.1: Task 2 (2 pts)

Paste the different views of the robot in a word document and then prepare a PDF document.

1.5.2 Task 3 (8 pts)

Make a copy of the sim.py file and make the necessary changes to solve the task. Save this file in this specific format <last*name*>_<*first_name*>.*py*.

1.5 Reference

https://mujoco.readthedocs.io/en/stable/APIreference/index.html